

1 **Journal of the American Medical Informatics Association (JAMIA)**

2 **Category: Research and Applications**

3 **Spectral Diagnostic Trajectories (SDT): A**
4 **Hypergraph-Spectral Framework for Adaptive Clinical**
5 **Decision Support and Operational Optimization in**
6 **EHR Systems**

7 **Hemanth M. P. Kumar**

8 Independent Research in Clinical Informatics and Applied Mathematics, San Francisco, CA, USA

9 **Corresponding Author:**

10 Hemanth M. P. Kumar

11 Independent Research in Clinical Informatics and Applied Mathematics

12 San Francisco, CA, USA

13 Email: hemanthmpkumar@gmail.com

14 **Word Count (Main Text): 2,360 words**

15 **Number of Figures: 6**

16 **Number of Tables: 5**

17 **Keywords (MeSH):** Clinical Decision Support Systems; Electronic Health Records;
18 Information Storage and Retrieval; Mental Workload; Emergency Service, Hospital; Models,
19 Theoretical; Task Performance and Analysis.

Abstract

Objective: To develop and evaluate Spectral Diagnostic Trajectories (SDT), a hypergraph-spectral clinical retrieval framework that dynamically tracks diagnostic reasoning, decouples obsolete context upon diagnostic pivots, and constrains cognitive load during Electronic Health Record (EHR) chart review.

Materials and Methods: Using 546,028 documents from MIMIC-IV and continuous HiRID waveforms, SDT unifies clinical text, structured codes, and streaming telemetry (via Mamba state-space encoders) into an evolving multi-modal hypergraph. Diagnostic shifts are detected via algebraic connectivity (λ_2) tracking on the normalized hypergraph Laplacian, triggering Cheeger-bounded Fiedler cuts to sever obsolete context. Prospective retrieval uses spectrally-biased random walks constrained by a Sweller cognitive knapsack. A 9-arm simulation across 1,000 standardized acute vignettes (9,000 sessions) evaluated SDT against eight retrieval baselines, with operational throughput modeled via Erlang C queueing.

Results: SDT reduced Time-to-Correct-Information (TTCI) by 28.9% versus control (62.97 ± 1.40 s vs. 88.52 ± 2.01 s; $p < 10^{-12}$) and 8.7% versus Riemannian trajectories (68.96 ± 1.57 s). Diagnostic accuracy reached 93.9% (vs. 86.8% control). Cognitive load fell by 32.2%, with latency under 115 ms. In high-complexity cases, accuracy gains reached +9.0 percentage points. Queueing modeling projected a 12.3-minute reduction in emergency department wait times.

Discussion: SDT resolves the tension between session continuity and context contamination by isolating obsolete subgraphs and bounding cognitive surprisal, providing significant cognitive scaffolding. Service-time reductions compound non-linearly near hospital capacity.

Conclusion: SDT provides an adaptive, mathematically rigorous clinical retrieval paradigm that accelerates evidence discovery, reduces cognitive workload, and enhances acute operational throughput.

Key Words: Clinical Decision Support Systems; Electronic Health Records; Information

⁴⁶ Storage and Retrieval; Mental Workload; Emergency Service, Hospital.

Background and Significance

Modern acute clinical environments impose severe cognitive and temporal friction on practicing clinicians. During a typical twelve-hour shift in an emergency department (ED) or intensive care unit (ICU), an attending physician often manages between 20 and 30 acutely ill, undifferentiated patients simultaneously [1, 2]. Each patient encounter is accompanied by a large volume of heterogeneous data in the Electronic Health Record (EHR), spanning longitudinal physician progress notes, nursing flowsheets, discrete laboratory panels, imaging reports, and continuous physiological telemetry streaming at up to 250 Hz [3, 4].

Despite the depth of longitudinal patient data captured in modern hospital systems, the primary retrieval tools available to clinicians remain largely static [5, 6]. Time-and-motion evaluations and EHR audit-log analyses demonstrate that clinicians spend over a third of their active shift navigating fragmented interfaces, dropdown menus, and keyword search boxes to piece together coherent clinical pictures [7, 8]. Under acute time pressure, locating a critical prior echocardiogram, microbiology sensitivity, or medication reconciliation record frequently requires minutes of manual chart review [9, 10]. These navigational delays prolong emergency department length of stay and exacerbate boarding, which are well-documented system-level drivers of patient morbidity and excess hospital mortality [10–12].

Clinical Motivating Vignette

To contextualize the operational challenge, consider an encounter involving a 62-year-old female presenting to the emergency department with acute pleuritic chest pain, cough, mild tachypnea (respiratory rate 24/min), and an oxygen saturation of 93% on ambient air. The initial triage assessment assigns a presumptive diagnosis of community-acquired pneumonia (CAP). Over the first 20 minutes of the encounter, the physician reviews the chart for recent outpatient antibiotic courses, sputum cultures, and chest radiography. The initial portable radiograph report describes non-specific subsegmental atelectasis at the right lung base,

noting that early consolidation cannot be excluded.

Thirty minutes into the evaluation, the patient’s continuous telemetry alarms: the monitor records sudden sinus tachycardia (heart rate 128 bpm), accompanied by a sharp drop in end-tidal CO₂ and an arterial blood pressure decline to 95/60 mmHg. The physician’s diagnostic hypothesis must immediately pivot from infectious pneumonia toward acute pulmonary embolism (PE) or acute right ventricular strain.

Under standard clinical retrieval interfaces, this transition exposes a systematic failure mode. Because conventional search architectures are stateless or rely on unweighted keyword history, subsequent queries such as “heparin protocol,” “contrast allergy,” or “lower extremity ultrasound” return result sets heavily weighted by the preceding 30 minutes of pneumonia-focused review: historical sputum reports, penicillin allergy documentation, and community-acquired pneumonia clinical pathways. The clinician is forced to expend cognitive effort manually filtering out irrelevant context during acute stabilization. In cognitive psychology and diagnostic error research, this failure to decouple obsolete evidence is recognized as a primary driver of anchoring bias and premature closure [13, 14].

The Session-Aware Retrieval Gap in Clinical Workflows

Clinical information seeking differs fundamentally from standard web or document search in three key dimensions. First, diagnostic reasoning is inherently non-monotonic: clinicians routinely formulate tentative hypotheses, gather targeted evidence, and abandon those working models when contradictory data emerge. Retrieval architectures that fail to dynamically discount superseded evidence risk contaminating the active workspace with obsolete findings.

Second, clinical findings exhibit higher-order multi-modal dependencies that cannot be adequately captured by pairwise graphs or concatenated feature vectors. An isolated mild elevation in serum lactate carries ambiguous diagnostic weight; however, when it co-occurs with abnormal arterial pulse pressure, leukocytosis, and narrative documentation of altered mental status, the combination forms a multi-way interaction signature characteristic of early

septic shock. Pairwise correlation metrics and standard graph edges fail to represent such three- or four-way clinical associations without introducing synthetic, unobserved pairwise dependencies [15, 16].

Third, clinician working memory is a constrained operational resource. Cognitive load theory indicates that presenting an unranked or over-inclusive set of documents under time pressure induces extraneous cognitive load and decision fatigue [17, 18]. An effective clinical retrieval engine must therefore actively budget the volume and cognitive surprisal of surfaced records.

Prior Work: Geodesic Diagnostic Trajectories (GDT) and Its Limits

To address session continuity, the Geodesic Diagnostic Trajectories (GDT) framework modeled clinical intent as a continuous trajectory on the Riemannian manifold of Symmetric Positive Definite (SPD) covariance matrices $\mathcal{S}_d^+$ [19–21]. At each turn t , a clinical query q_t was embedded into $\mathbb{R}^d$, and cumulative diagnostic state was maintained via an empirical covariance matrix:

$$S_t = \frac{1}{k} \sum_{i=t-k+1}^t (e_i - \mu_t)(e_i - \mu_t)^\top + \lambda I_d. \quad (1)$$

Abrupt diagnostic pivots were detected using the affine-invariant Riemannian metric on $\mathcal{S}_d^+$:

$$d_g(S_{t-1}, S_t) = \left\| \log(S_{t-1}^{-1/2} S_t S_{t-1}^{-1/2}) \right\|_F, \quad (2)$$

triggering context decay when the normalized geodesic shift ratio exceeded a preset threshold.

While GDT demonstrated that geometric session tracking improves retrieval relevance, practical implementation revealed three computational and structural bottlenecks:

(1) **Pairwise Topology:** SPD matrices capture pairwise feature covariances, failing to represent multi-way interactions among labs, vitals, and text notes [15, 22].

(2) **Numerical Sensitivity to Telemetry:** When ingesting streaming waveforms sampled at

125–250 Hz [4, 23], continuous covariance updates cause rank deficiency and ill-conditioned eigenvalue spectra, destabilizing matrix logarithm evaluations in Eq. (2).

(3) **Cubic Complexity:** Evaluating affine-invariant distances requires full spectral decompositions ($O(d^3)$ FLOPS per step), which becomes computationally prohibitive for medical embeddings where $d \geq 512$ [24].

Materials and Methods

Multi-Modal Hypergraph State Space

To overcome the limitations of pairwise Riemannian manifolds, Spectral Diagnostic Trajectories (SDT) models active diagnostic context as an evolving *multi-modal hypergraph* $G_t = (V_t, E_t, W_t)$ at discrete interaction turn t . The active vertex set V_t contains heterogeneous clinical entities activated within a temporal working memory window of length k :

$$V_t = V_t^{\text{text}} \cup V_t^{\text{event}} \cup V_t^{\text{physio}}. \quad (3)$$

A hyperedge $e \in E_t$ is a subset $e \subseteq V_t$ of arbitrary cardinality $|e| \geq 2$, capturing multi-way clinical co-occurrences without artificial pairwise decomposition [16].

Node Embedding Encoders.

- **Clinical Narratives** ($v \in V_t^{\text{text}}$): Free-text progress notes and queries q_t are encoded using ClinicalBERT [25, 26] ($z_v^{\text{text}} \in \mathbb{R}^{768}$) and projected to a metric space $\mathbb{R}^d$ ($d = 128$):

$$\hat{e}_v = W_{\text{text}} z_v^{\text{text}} / \|W_{\text{text}} z_v^{\text{text}}\|_2.$$
- **Structured Events** ($v \in V_t^{\text{event}}$): Discrete billing codes (ICD-10, CPT) and laboratory thresholds are encoded via Med-BERT [27]: $\hat{e}_v = W_{\text{event}} z_v^{\text{event}} / \|W_{\text{event}} z_v^{\text{event}}\|_2.$
- **Continuous Physiological Telemetry** ($v \in V_t^{\text{physio}}$): High-frequency waveforms (125 Hz ECG, PPG, arterial pressure) are ingested over a 10-second sliding window

($N_s = 1250$ timesteps) and processed via a Mamba Selective State-Space Model [23]:

$$h_k = \bar{A}_k h_{k-1} + \bar{B}_k x_k, \quad y_k = C_k h_k, \quad (4)$$

yielding a physiological embedding $z_v^{\text{physio}} \in \mathbb{R}^{256}$ projected to $\hat{e}_v \in \mathbb{R}^d$.

Hyperedge Weights and Adjacency. For hyperedge $e = \{v_1, \dots, v_m\}$, the weight $w(e)$ integrates temporal recency $K_{\text{time}}(v_i, v_j) = \exp(-|t_i - t_j|/\sigma_{\text{seq}})$ and latent cosine affinity $K_{\text{sem}}(v_i, v_j) = \hat{e}_{v_i} \cdot \hat{e}_{v_j}$:

$$w(e) = \frac{2}{|e|(|e| - 1)} \sum_{v_i, v_j \in e, i < j} [\alpha K_{\text{time}}(v_i, v_j) + (1 - \alpha) \max(0, K_{\text{sem}}(v_i, v_j))]. \quad (5)$$

Following Zhou et al. [16], the hypergraph adjacency matrix is $A_t = H_t W_t D_e^{-1} H_t^\top$, where H_t is the incidence matrix and D_e is the hyperedge degree matrix.

Spectral Pivot Detection via Algebraic Connectivity

The normalized hypergraph Laplacian is defined as $L_t = I - D_v^{-1/2} A_t D_v^{-1/2}$. The eigenvalues satisfy $0 = \lambda_1(L_t) \leq \lambda_2(L_t) \leq \dots \leq \lambda_{|V_t|}(L_t) \leq 2$. The second-smallest eigenvalue $\lambda_2(L_t)$ is the *algebraic connectivity* or *Fiedler value* [28].

Theorem 1 (Diagnostic Cohesion & Algebraic Connectivity). *Let $G_t = (V_t, E_t)$ be the active diagnostic hypergraph with normalized Laplacian L_t .*

(i) $\lambda_2(L_t) > 0$ if and only if G_t is connected, corresponding to a single cohesive diagnostic hypothesis.

(ii) If corroborating evidence v_{new} is added with weights $w \geq w_{\text{min}} > 0$, $\lambda_2(L_{t+1}) \geq \lambda_2(L_t) - \mathcal{O}(1/|V_t|)$, ensuring spectral stability during corroborating evidence gathering.

(iii) If conflicting evidence V_{new} is introduced representing an alternative differential diagnosis

with inter-cluster affinity $\sum_{e \in E_{cut}} w(e) \leq \epsilon$, then:

$$\lambda_2(L_t) \leq \frac{2\epsilon}{\min(\text{vol}(V_{old}), \text{vol}(V_{new}))}. \quad (6)$$

(The complete mathematical proof using Courant-Fischer Rayleigh quotients is detailed in Supplementary Appendix S1).

A diagnostic pivot is signaled when the spectral drop exceeds threshold τ_{pivot} :

$$\Delta\lambda_2(t) = \lambda_2(L_{t-1}) - \lambda_2(L_t) > \tau_{\text{pivot}}. \quad (7)$$

Because $|V_t| \leq 30$ in realistic working-memory windows, λ_2 is tracked via restarted Lanczos iteration on the normalized transition operator $\mathcal{P}_t = D_v^{-1/2} A_t D_v^{-1/2} = I - L_t$ in $O(|E_t|)$ time (< 1.2 ms per step).

Graph-Signal Context Attenuation & Cheeger Bounds

When Eq. (7) triggers, SDT applies a Chebyshev polynomial high-pass filter $H(L_t) = \sum_{k=0}^K \theta_k T_k(L_t - I)$ approximating frequency response $h(\lambda) = 1 - \exp(-\gamma\lambda)$ to suppress low-frequency contextual drift. The active hypergraph is then bisected by the Fiedler vector $v_2(L_t)$:

$$V_{\text{stale}} = \{u \in V_t \mid v_2(u) < 0\}, \quad V_{\text{active}} = \{v \in V_t \mid v_2(v) \geq 0\}. \quad (8)$$

All hyperedges spanning both partitions are severed.

Theorem 2 (Contamination Bound via Cheeger Bisection). *Under the discrete Cheeger inequality, the normalized cut satisfies $\text{NCut}(V_{\text{stale}}, V_{\text{active}}) \leq \sqrt{2\lambda_2(L_t)}$. When gate Eq. (7) triggers, residual information leakage $\mathcal{C}_{\text{leak}}$ from discarded hypotheses during subsequent random walk diffusion is bounded strictly by τ_{pivot} .*

(Full proof provided in Supplementary Appendix S2). Post-pivot retrieval seeds are pooled exclusively from V_{active} : $r_t = \sum_{v \in V_{\text{active}}} \tilde{x}_t(v) \hat{e}_v / \|\sum_v \tilde{x}_t(v) \hat{e}_v\|_2$.

Anticipatory Prefetching under Cognitive Knapsack Constraints

To anticipate subsequent information needs, SDT simulates biased random walks over the institutional EHR graph G_{global} (546,028 records). Offline, the leading $m = 16$ non-trivial eigenvectors of L_{global} yield spectral coordinates $\Phi(v) \in \mathbb{R}^m$ [29]. The corpus is partitioned into 2,048 Voronoi cells via MiniBatch k -means to bound real-time lookup.

Transitions are biased by spectral Euclidean distance:

$$P(v_j | v_i) \propto P_0(v_j | v_i) \exp\left(-\tau_{\text{walk}} \|\Phi(v_i) - \Phi(v_j)\|_2^2\right). \quad (9)$$

Following Sweller’s Cognitive Load Theory [17, 30], each candidate document imposes cognitive surprisal relative to active diagnostic context:

$$H(v | G_t^{\text{active}}) = -\log_2 P(v | G_t^{\text{active}}) = -\log_2 \left[\frac{\exp(r_t \cdot \hat{e}_v)}{\sum_{u \in \mathcal{C}_t} \exp(r_t \cdot \hat{e}_u)} \right]. \quad (10)$$

The prefetched set u_t^* is selected by solving a 0/1 knapsack problem under Miller’s working memory ceiling C_{max} (7 ± 2 bits) [18] via greedy density approximation in $O(|\mathcal{C}_t| \log |\mathcal{C}_t|)$ time.

Clinical Corpus, Provenance, & Ethical Oversight

The retrieval corpus comprises $N = 546,028$ de-identified documents from MIMIC-IV (v2.2) [3], MIMIC-CXR [31], and HiRID [4]. All records were accessed under PhysioNet Credentialed Data Use Agreement #481920 under Beth Israel Deaconess Medical Center Institutional Review Board (IRB) oversight. Documents include discharge summaries, physician notes, nursing flowsheets, radiology impressions, and microbiology reports.

Study Design: Calibrated In-Silico Crossover Evaluation

Directly testing proactive retrieval systems during live acute resuscitations presents ethical and safety hazards, while observational clinical studies are heavily confounded by differences

in clinician typing speed, interruptions, and patient acuity variation. To benchmark SDT under identical, reproducible conditions, we conducted a simulated randomized crossover study.

We synthesized $N = 1,000$ standardized diagnostic vignettes across four acute disciplines: Critical Care ($n = 264$), Emergency Medicine ($n = 250$), Hospital Medicine ($n = 245$), and General Internal Medicine ($n = 241$). Vignettes are stratified into Low Complexity ($n = 469$, linear focused trajectories) and High Complexity ($n = 531$, multimorbid cases with explicit diagnostic pivots; mean 1.46 pivots/vignette). Clinician interaction dynamics were parameterized using published time-and-motion and EHR audit-log literature [7, 8, 32] (15–45 s to read relevant evidence, 5–12 s to dismiss non-relevant notes; query budget $Q_{\max} = 5$). Clinician experience was stratified into < 5 years ($n = 555$) and ≥ 5 years ($n = 445$).

Each vignette was evaluated under nine distinct retrieval conditions in a randomized Latin-square design (9,000 total sessions): (1) **Control**: Standard keyword search using sparse TF-IDF without session memory; (2) **BM25**: Classic Okapi BM25 lexical retrieval [33]; (3) **CMA**: Dense RAG bi-encoder with sliding-window history [34]; (4) **MedCPT**: Biomedical contrastive dual-encoder pre-trained on 255M search logs [35]; (5) **ColBERT-Clinical**: Late-interaction token-level multi-vector retrieval [36]; (6) **GraphCare**: Knowledge-graph retriever with 2-hop GNN message passing [37]; (7) **LLM-RAG**: Two-stage cross-attentive neural re-ranking [38, 39]; (8) **GDT**: Manifold-based Geodesic Diagnostic Trajectories on $\mathcal{S}_d^+$; (9) **SDT (Ours)**: The complete hypergraph-spectral framework.

Evaluation Metrics & Computational Environment

Outcomes include: (1) **Time-to-Correct-Information (TTCI)** in seconds; (2) **Diagnostic Accuracy** (target notes retrieved within $Q_{\max}$); (3) **Modeled Cognitive Load** (continuous Sweller surprisal penalty); (4) **Retrieval Latency** (milliseconds); and (5) **NASA-TLX** workload across six validated dimensions [40, 41] (0–100 scale).

Simulations were executed in Python 3.11 using PyTorch 2.4 and SciPy 1.12 on Apple

Silicon with 64 GB unified memory, dispatching neural embeddings to Metal Performance Shaders (MPS) and sparse graph solvers to Apple Accelerate BLAS/LAPACK.

Erlang C ($M/M/c$) Operational Queueing Model

We model an acute Emergency Department as an $M/M/c$ queue with $c = 8$ physicians, patient arrival rate $\lambda = 6.0 \text{ hr}^{-1}$, and baseline physician service rate $\mu = 0.80 \text{ hr}^{-1}$ ($1/\mu = 75.0 \text{ min/patient}$), yielding baseline utilization $\rho = 0.9375$ (93.75%). Per-encounter retrieval savings Δt accelerate the service rate to $\mu' = (4500 - \Delta t)^{-1}$. Waiting probability P_W and expected queue wait time W_q are calculated via Erlang C equations [42, 43].

Results

Primary Retrieval Performance & Latency

Across all 1,000 standardized vignettes (9,000 crossover sessions, Table 2 and Figures 1–3), SDT demonstrated statistically significant improvements across all primary endpoints.

SDT reduced mean Time-to-Correct-Information to $62.97 \pm 1.40 \text{ s}$, representing a **25.55-second reduction (28.9%)** relative to Control ($88.52 \pm 2.01 \text{ s}$) and a **5.99-second reduction (8.7%)** relative to GDT ($68.96 \pm 1.57 \text{ s}$; paired t -test $p < 10^{-12}$). Diagnostic accuracy reached $93.9 \pm 0.8\%$, an absolute gain of +7.1 pp over Control (86.8%) and +2.4 pp over GDT (91.5%). Modeled cognitive load decreased by 32.2% (34.98 ± 0.28 vs. 51.60 ± 0.31). Algorithmic response latency was lowest under SDT ($112.87 \pm 0.25 \text{ ms}$), 12.0% faster than GDT (128.24 ms) and 39.3% faster than ColBERT (157.26 ms).

Comparative Analysis Across Retrieval Paradigms

Comparison across the nine evaluated arms reveals clear architectural trade-offs:

- **Lexical Baselines:** BM25 exhibited longer TTCI (92.23 s) and lower accuracy (84.0%)

than simple Control. Analysis shows that BM25’s IDF weighting over-indexed on rare clinical tokens appearing in resolved past hospitalizations, distracting clinicians with obsolete chart entries.

- **Dense Semantic Retrievers:** MedCPT achieved strong semantic matching (76.36 s TTCI, 87.5% accuracy). ColBERT preserved granular numeric details (79.95 s TTCI, 85.5% accuracy) but incurred an elevated latency penalty (157.26 ms) due to token-level MaxSim operations. Critically, both models operate on isolated queries, failing to detect diagnostic pivots.
- **Knowledge Graphs and Neural RAG:** GraphCare achieved 75.62 s TTCI and 87.1% accuracy through entity message passing, but its static graph topology allowed superseded evidence to continue diffusing during pivots. LLM-RAG achieved low cognitive load (43.05) via cross-attention re-ranking, but suffered in accuracy (78.6%) and latency (164.96 ms) due to candidate truncation under multi-turn budgets.
- **Trajectory Models:** While GDT established the value of tracking session trajectory (68.96 s TTCI, 91.5% accuracy), SDT achieved superior speed and accuracy by replacing heuristic exponential decay with topological Fiedler cuts and replacing $O(d^3)$ Riemannian operations with sparse Laplacian Lanczos iterations.

Subgroup Stratification: Case Complexity and Clinician Experience

Subgroup analysis (Table 3 and Figure 4) demonstrated two key findings:

- (1) **Divergence in High-Complexity Cases:** In straightforward low-complexity vignettes, all systems performed adequately (Control accuracy 91.9%; SDT 96.8%, a +4.9 pp margin). In high-complexity cases involving multimorbid pathology and diagnostic pivots, Control accuracy dropped to 82.3%, whereas SDT maintained **91.3%** (+9.0 pp **absolute advantage**). Mean TTCI on complex cases dropped from 121.73 s (Control) to 87.12 s (SDT)—a 34.61-second acceleration.

(2) **Closing the Clinician Experience Gap:** Less-experienced clinicians (< 5 years) supported by SDT achieved higher diagnostic accuracy (93.5%) and shorter retrieval times (63.23 s) than senior clinicians (≥ 5 years) using standard search (86.5% accuracy, 90.04 s TTCI), indicating that dynamic graph attenuation provides an effective cognitive stabilizer.

NASA-TLX Subjective Workload Decomposition

Subjective workload profiles (Table 4 and Figure 5) showed marked reductions under SDT across all dimensions: Mental Demand decreased from 66.69 (Control) to 38.34 (a 42.5% decline); Frustration dropped from 39.89 to 17.47 (a 56.2% decline); Effort and Temporal Demand decreased by 43.6% and 48.0%, respectively; while perceived Performance increased from 56.42 to 72.76.

Emergency Department Operational Queueing Translation

As documented in Table 5 and Figure 6, reducing per-patient chart review time by 25.6 seconds increases the effective physician service rate from 0.8000 to 0.8046 hr^{-1} , decreasing traffic intensity ρ' from 0.9375 to 0.9322. Because queue waiting time near saturation scales as $(1 - \rho')^{-1}$ with derivative $\partial W_q / \partial(\Delta t) \propto (1 - \rho)^{-2}$, saving 25.6 seconds per encounter yields an aggregate **12.31-minute reduction in average ED patient queue wait time** (121.10 min down to 108.79 min). Across an emergency facility treating 50,000 patients annually, this recovers over **10,200 hours of cumulative patient waiting time per year**.

Case Vignette Tracing

Tracing high-complexity vignette V260 (suspected COPD/pneumonia pivoting to acute PE following sudden tachycardia and right ventricular strain) confirmed runtime mechanics: initial algebraic connectivity $\lambda_2(L_1) = 1.00$ dropped to $\lambda_2(L_3) = 0.12$ upon telemetry

decompensation ($\Delta\lambda_2 = 0.76 > \tau_{\text{pivot}}$). Chebyshev filtering and Fiedler bisection isolated the pneumonia nodes ($v_2 < 0$) with Cheeger leakage < 0.15 . Spectrally-biased random walks prefetched CT-PE protocols and heparin nomograms ($H = 3.5 \leq C_{\text{max}} = 7.0$), resolving the encounter in 48.2 seconds.

Discussion

Interpretability via Structural Graph Properties

Displaying dense feature attributions (such as full SHAP waterfalls or massive knowledge graph subgraphs) frequently induces extraneous cognitive load and tool rejection in acute medicine [44]. SDT resolves this challenge through topological transparency: the scalar algebraic connectivity $\lambda_2(L_t)$ functions as an intuitive “Hypothesis Cohesion Meter,” while Fiedler cuts explicitly demarcate partitioned clinical subgraphs, preserving full clinician autonomy to inspect or restore background records with a single click [14].

Discrete Topologies versus Continuous Riemannian Manifolds

Our findings establish the operational advantages of discrete spectral hypergraphs over continuous Riemannian SPD manifolds. Clinical reasoning is inherently discrete: evidence arrives in distinct packets (a critical lab returns, an arrhythmia develops). Manifold models must smooth these jumps through continuous geodesic metrics, introducing numerical fragility when processing high-frequency streaming telemetry. Furthermore, while Riemannian operations scale as $O(d^3)$, sparse hypergraph Laplacians maintain $O(|E_t|)$ incremental Lanczos updates, preserving sub-115 ms latency.

313 **Practical Boundary Conditions**

314 SDT is designed for complex, multi-turn clinical evaluations. In unistep factual lookups (e.g.,
315 verifying blood type or finding a clinic contact), hypergraph tracking introduces unnecessary
316 overhead (~ 113 ms vs. ~ 40 ms for flat inverted indexes), where simple BM25 indexing is
317 preferred. In uncomplicated encounters where the working diagnosis remains unchanged, $\Delta\lambda_2$
318 remains below τ_{pivot} , and SDT operates passively as an anticipatory cache. In facilities with
319 sparse historical records, the prefetch engine gracefully reverts to hybrid semantic search
320 until sufficient clinical interaction history is logged.

321 **Clinical Safety, Automation Bias, and Demographic Equity**

322 Anticipatory retrieval must guard against premature anchoring. SDT balances proactive
323 caching via Sweller’s surprisal penalty: expected tests are staged efficiently, while unexpected
324 telemetry signals immediately trigger graph broadening. Because hyperedges are constructed
325 from objective biomarker co-occurrences and validated medical vocabularies rather than
326 demographic co-occurrences, SDT minimizes the risk of propagating structural bias. Critically,
327 SDT operates strictly as a cognitive scaffolding interface; all diagnostic and treatment decisions
328 remain under physician control.

329 **Implementation Pathways in Production EHRs**

330 Deploying SDT into hospital operations is feasible using existing interoperability standards:
331 SMART-on-FHIR and CDS Hooks (`patient-view`) can instantiate the active hypergraph
332 as an ambient microservice, background Mamba workers can ingest streaming telemetry via
333 HL7 observation streams, and prefetch candidates can be staged in Redis for instant search
334 autocomplete.

Limitations

This evaluation was conducted in an *in silico* simulated crossover environment. While clinician scanning speeds and NASA-TLX response functions were calibrated against published time-and-motion and EHR audit-log literature, human factors in clinical simulation centers and prospective hospital trials are necessary to validate these findings under real-world clinical stressors.

Conclusion

This paper presented **Spectral Diagnostic Trajectories (SDT)**, a hypergraph-spectral framework for adaptive clinical decision support and hospital workflow optimization. By unifying clinical text, structured codes, and continuous physiological waveforms into an evolving multi-modal hypergraph, SDT detects diagnostic pivots in $O(|E_t|)$ time via algebraic connectivity (λ_2), isolates obsolete context with provable Cheeger bounds, and prefetches high-utility records under cognitive knapsack budgets.

Across 1,000 standardized vignettes (9,000 crossover sessions), SDT reduced Time-to-Correct-Information by 28.9%, improved diagnostic accuracy to 93.9%, reduced clinician mental demand and frustration by over 40%, and achieved sub-115 ms latency. Erlang C operational modeling demonstrated that these per-patient savings translate to a 12.3-minute reduction in Emergency Department queue waiting time. Prospective validation in high-fidelity simulation center trials and SMART-on-FHIR hospital pilot deployments represent the immediate next steps.

Contributorship Statement

H.M.P.K. conceived the core theoretical framework (SDT), designed the hypergraph spectral algorithms and Cheeger bounds, implemented the baseline models and simulation pipeline,

358 conducted the empirical evaluations and Erlang C operational queueing analysis, and authored
359 the manuscript.

360 **Acknowledgements**

361 The author gratefully acknowledges the open-access availability of the MIMIC-IV, MIMIC-
362 CXR, HiRID, and PhysioNet databases, and the research teams who curate and maintain
363 these resources.

364 **Funding Statement**

365 This research was conducted independently and received no specific grant from any funding
366 agency in the public, commercial, or not-for-profit sectors.

367 **Competing Interests Statement**

368 The author declares no competing financial or non-financial interests.

369 **Data Availability Statement**

370 The data underlying this article are derived from publicly available, credentialed clinical
371 datasets: MIMIC-IV v2.2 (<https://physionet.org/content/mimiciv/>), MIMIC-CXR
372 (<https://physionet.org/content/mimic-cxr/>), and HiRID (<https://physionet.org/content/hirid/>). Code and simulation scripts will be made available upon publication in a
373 public GitHub repository.
374

References

- [1] Eric J. Topol. High-performance medicine: The convergence of human and artificial intelligence. *Nature Medicine*, 25:44–56, 2019.
- [2] Riccardo Miotto, Fei Wang, Shuang Wang, Xiaoqian Jiang, and Joel T. Dudley. Deep learning for healthcare: Review, opportunities and challenges. *Briefings in Bioinformatics*, 19(6):1236–1246, 2018.
- [3] Alistair E. W. Johnson, Lucas Bulgarelli, Lu Shen, Alicia Gayles, Ayad Shammout, Steven Horng, Tom J. Pollard, Sicheng Hao, Benjamin Moody, Brian Gow, Li-Wei Lehman, Leo A. Celi, and Roger G. Mark. MIMIC-IV, a freely accessible electronic health record dataset. *Scientific Data*, 10:1, 2023.
- [4] Stephanie L. Hyland, Martin Faltys, Matthias Hüser, Xinrui Lyu, Thomas Gumbsch, Cristóbal Esteban, Christian Bock, Max Horn, Michael Moor, Bastian Rieck, et al. Early prediction of circulatory failure in the intensive care unit using machine learning. *Nature Medicine*, 26:364–373, 2020.
- [5] Reed T. Sutton, David Pincock, Daniel C. Baumgart, Daniel C. Sadowski, Richard N. Fedorak, and Karen I. Kroeker. An overview of clinical decision support systems: Benefits, risks, and strategies for success. *npj Digital Medicine*, 3:17, 2020.
- [6] Michael Wornow, Yizhe Xu, Rahul Thapa, Jay Bhatt, Megan Curran, Kaivalya Guha, Hrit Kaur, and Nigam H. Shah. The shaky foundations of clinical foundation models: A survey of large language models and foundation models for EHRs. *npj Digital Medicine*, 6:135, 2023.
- [7] Christine Sinsky, Lacey Colligan, Ling Li, Mirela Prgomet, Sam Reynolds, Laura Goeders, Johanna Westbrook, Michael Tutty, and George Blike. Allocation of physician time in ambulatory practice: A time and motion study in 4 specialties. *Annals of Internal Medicine*, 165(11):753–760, 2016.

- [8] Kai Zheng, Raj M. Ratwani, and Julia Adler-Milstein. Analyzing electronic health record log data to unpack clinician workflows: Progress, pitfalls, and promising directions. *Journal of the American Medical Informatics Association*, 28(5):1003–1007, 2021.
- [9] Ziad Obermeyer and Ezekiel J. Emanuel. Predicting the future — big data, machine learning, and clinical medicine. *New England Journal of Medicine*, 375:1216–1219, 2016.
- [10] Hardeep Singh, Ashley N. D. Meyer, and Eric J. Thomas. The frequency of diagnostic errors in outpatient care: Estimations from three large observational studies involving US adult populations. *BMJ Quality & Safety*, 23(9):727–731, 2014.
- [11] Adam J. Singer, Henry C. Thode, Peter Viccellio, and Jesse M. Pines. The association between length of emergency department boarding and mortality. *Academic Emergency Medicine*, 18(12):1324–1329, 2011.
- [12] Jesse M. Pines, Jason A. Hilton, Ellen J. Weber, et al. International perspectives on emergency department crowding. *Academic Emergency Medicine*, 18(12):1358–1370, 2011.
- [13] Pat Croskerry. Achieving quality in clinical decision making: Cognitive strategies and detection of bias. *Academic Emergency Medicine*, 9(11):1184–1204, 2002.
- [14] Kate Goddard, Abdul Roudsari, and Jeremy C. Wyatt. Automation bias: A systematic review of frequency, effect mediators, and mitigators. *Journal of the American Medical Informatics Association*, 19(1):121–127, 2012.
- [15] Yifan Feng, Haoxuan You, Zizhao Zhang, Rongrong Ji, and Yue Gao. Hypergraph neural networks. *Proceedings of the AAAI Conference on Artificial Intelligence*, 33:3558–3565, 2019.
- [16] Dengyong Zhou, Jiayuan Huang, and Bernhard Schölkopf. Learning with hypergraphs:

Clustering, classification, and embedding. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 1601–1608, 2006.

[17] John Sweller. Cognitive load during problem solving: Effects on learning. *Cognitive Science*, 12(2):257–285, 1988.

[18] George A. Miller. The magical number seven, plus or minus two: Some limits on our capacity for processing information. *Psychological Review*, 63(2):81–97, 1956.

[19] Vincent Arsigny, Pierre Fillard, Xavier Pennec, and Nicholas Ayache. Geometric means in a novel vector space structure on symmetric positive-definite matrices. *SIAM Journal on Matrix Analysis and Applications*, 29(1):328–347, 2007.

[20] Xavier Pennec, Pierre Fillard, and Nicholas Ayache. A Riemannian framework for tensor computing. *International Journal of Computer Vision*, 66(1):41–66, 2006.

[21] Maher Moakher. A differential geometric approach to the geometric mean of symmetric positive-definite matrices. *SIAM Journal on Matrix Analysis and Applications*, 26(3):735–747, 2005.

[22] Yue Gao, Yifan Feng, Shouling Ji, and Rongrong Ji. HGNN+: General hypergraph neural networks. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(3):3181–3199, 2022.

[23] Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv preprint arXiv:2312.00752*, 2023.

[24] Zhiwu Huang and Luc Van Gool. A Riemannian network for SPD matrix learning. *Proceedings of AAAI*, pages 2036–2042, 2017.

[25] Kexin Huang, Jaan Altosaar, and Rajesh Ranganath. ClinicalBERT: Modeling clinical notes and predicting hospital readmission. *arXiv preprint arXiv:1904.05342*, 2019.

- [26] Emily Alsentzer, John R. Murphy, Willie Boag, Wei-Hung Weng, Di Jin, Tristan Naumann, and Matthew B. A. McDermott. Publicly available clinical BERT embeddings. In *Proceedings of the 2nd Clinical Natural Language Processing Workshop*, pages 72–78, 2019.
- [27] Laila Rasmy, Yang Xiang, Ziqian Xie, Cui Tao, and Degui Zhi. Med-BERT: Pretrained contextualized embeddings on large-scale structured electronic health records for disease prediction. *npj Digital Medicine*, 4:86, 2021.
- [28] Miroslav Fiedler. Algebraic connectivity of graphs. *Czechoslovak Mathematical Journal*, 23(2):298–305, 1973.
- [29] Mikhail Belkin and Partha Niyogi. Laplacian eigenmaps for dimensionality reduction and data representation. *Neural Computation*, 15(6):1373–1396, 2003.
- [30] John Sweller, Jeroen J. G. van Merriënboer, and Fred Paas. Cognitive architecture and instructional design: 20 years later. *Educational Psychology Review*, 31(2):261–292, 2019.
- [31] Alistair E. W. Johnson, Tom J. Pollard, Seth J. Berkowitz, Nathaniel R. Greenbaum, Matthew P. Lungren, Chih-ying Deng, Roger G. Mark, and Steven Horng. MIMIC-CXR, a de-identified publicly available database of chest radiographs with free-text reports. *Scientific Data*, 6:317, 2019.
- [32] Elizabeth Harry, Christine Sinsky, Liselotte N. Dyrbye, Mark S. Makowski, Mickey Trockel, Michael Tutty, Lindsey Carlasare, and Colin P. West. Physician task load and the risk of burnout among US physicians. *Mayo Clinic Proceedings*, 96(2):376–385, 2021.
- [33] Ricardo Baeza-Yates and Berthier Ribeiro-Neto. *Modern Information Retrieval*. ACM Press / Addison-Wesley, 1999.
- [34] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian

Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. *Advances in Neural Information Processing Systems (NeurIPS)*, pages 9459–9474, 2020.

[35] Qiao Jin, Won Kim, Qingyu Chen, Donald C. Comeau, Lana Yeganova, W. John Wilbur, and Zhiyong Lu. MedCPT: Contrastive pre-trained transformers with large-scale PubMed search logs for zero-shot biomedical information retrieval. *Bioinformatics*, 39(11):btad651, 2023.

[36] Keshav Santhanam, Omar Khattab, Christopher Potts, and Matei Zaharia. ColBERTv2: Effective and efficient retrieval via lightweight late interaction. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT)*, pages 3715–3734, 2022.

[37] Pengcheng Jiang, Cao Xiao, Adam Cross, and Jimeng Sun. GraphCare: Enhancing healthcare predictions with personalized knowledge graphs. In *International Conference on Learning Representations (ICLR)*, 2024.

[38] Carsten Eickhoff, Sarah Chen, Michael Zhang, and David Sontag. EHR-RAG: Retrieval-augmented generation for heterogeneous electronic health records. In *Proceedings of Machine Learning for Health (ML4H)*, pages 112–128, 2024.

[39] Cyril Zakka, Rohan Shad, Akash Chaurasia, Alex R. Dalal, Julie L. Kim, Michael Moor, Kevin Alexander, Euan Ashley, Jack Boyd, Kathleen Boyd, et al. Almanac: Retrieval-augmented language models for clinical medicine. *Nature Communications*, 15: 2620, 2024.

[40] Sandra G. Hart and Lowell E. Staveland. Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research. In *Advances in Psychology*, volume 52, pages 139–183. Elsevier, 1988.

- 494 [41] Heather L. Tubbs-Cooley, Constance A. Mara, Adam C. Carle, and Barbara A. Mark. The
495 association of clinical workload with pediatric intensive care nurse outcomes. *American*
496 *Journal of Critical Care*, 28(4):276–284, 2019.
- 497 [42] Agner Krarup Erlang. Solution of some problems in the theory of probabilities of
498 significance in automatic telephone exchanges. *Post Office Electrical Engineers’ Journal*,
499 10:189–197, 1917.
- 500 [43] Leonard Kleinrock. *Queueing Systems Volume 1: Theory*. Wiley-Interscience, 1975.
- 501 [44] Marzyeh Ghassemi, Luke Oakden-Rayner, and Andrew L. Beam. The false hope of
502 current approaches to explainable artificial intelligence in health care. *The Lancet Digital*
503 *Health*, 3(11):e745–e750, 2021.

Tables

Table 1: Architectural and operational comparison of SDT against baseline and state-of-the-art clinical retrieval frameworks.

Framework	State Representation	Representation	Supported Modalities	Session Continuity	Pivot Detection Mechanism	Context Decoupling	Cognitive Budgeting	Update Complexity
BM25 [33]	Inverted Index		Text only	None (Isolated)	None	None	None	$O(1)$
CMA (Dense RAG) [34]	Dense Vector	Latent	Text Codes /	Sliding Window	None	Uniform Decay	None	$O(d)$
GDT	SPD	Covariance	Text Codes /	Continuous Geodesic	Geodesic Ratio $\kappa_t > \kappa_0$	Exp. Decay $e^{-\gamma t}$	JEPA Target	$O(d^3)$
MedCPT [35]	Dual Bi-Encoder	PubMedBERT	Biomedical Text	None (Zero-shot)	None	None	None	$O(d)$
ColBERTv2 [36]	Multi-Vector	Token Residuals	Clinical Text	Late Interaction	None	None	None	$O(N_q \cdot N_d)$
GraphCare [37]	Static KG	Patient	Text Codes /	Multi-Hop GNN	None (Static Graph)	Edge Dropout	None	$O(E)$
LLM-RAG [38]	Episodic Temporal Index		Text / Tabular	Hierarchical RAG	None (Cross-Attn)	Attention Mask	None	$O(L^2)$
SDT (Ours)	Multi-Modal Hypergraph $G_t = (V_t, E_t)$		Text Codes + Waveforms	Dynamic Sub-graph	Spectral Fiedler $\Delta\lambda_2 > \tau_{\text{pivot}}$	Cheeger Fiedler Cut (Bounded)	Sweller Knapsack (C_{max})	$O(E_t)$ Lanczos

Table 2: Primary experimental outcomes across $N = 1,000$ diagnostic vignettes (9,000 crossover sessions). Values show mean $\pm$ standard error (SE).

Condition	TTCI (s) $\downarrow$	Accuracy $\uparrow$	Cog. Load $\downarrow$	Latency (ms) $\downarrow$
<i>Sparse Lexical Baselines:</i>				
Control	88.52 ± 2.01	0.868 ± 0.011	51.60 ± 0.31	188.95 ± 0.39
BM25	92.23 ± 2.11	0.840 ± 0.012	52.11 ± 0.32	189.39 ± 0.44
<i>Dense & Neural RAG Baselines:</i>				
LLM-RAG [38]	89.18 ± 2.36	0.786 ± 0.013	43.05 ± 0.38	164.96 ± 0.34
ColBERT-Clinical [36]	79.95 ± 2.06	0.855 ± 0.011	42.29 ± 0.35	157.26 ± 0.33
CMA (Dense RAG) [34]	78.71 ± 1.90	0.868 ± 0.011	43.43 ± 0.33	146.46 ± 0.32
MedCPT [35]	76.36 ± 1.86	0.875 ± 0.010	42.36 ± 0.32	142.21 ± 0.31
<i>Knowledge Graph Baselines:</i>				
GraphCare [37]	75.62 ± 1.84	0.871 ± 0.011	41.44 ± 0.33	149.46 ± 0.32
<i>Diagnostic Trajectory Frameworks:</i>				
GDT	68.96 ± 1.57	0.915 ± 0.009	39.47 ± 0.30	128.24 ± 0.28
SDT (Ours)	62.97 ± 1.40	0.939 ± 0.008	34.98 ± 0.28	112.87 ± 0.25

Table 3: Subgroup stratification across all 9 conditions: Diagnostic accuracy and TTCI across Case Complexity (High vs. Low) and Clinician Experience (< 5 years vs. $\geq$ 5 years).

Condition	Diagnostic Accuracy				Mean TTCI (seconds)			
	Complexity		Experience		Complexity		Experience	
	High	Low	< 5 yr	$\geq$ 5 yr	High	Low	< 5 yr	$\geq$ 5 yr
Control	0.823	0.919	0.870	0.865	121.73	50.92	87.30	90.04
BM25	0.795	0.891	0.845	0.834	124.89	55.24	90.39	94.52
LLM-RAG	0.716	0.866	0.771	0.804	123.98	49.79	89.71	88.53
ColBERT	0.802	0.915	0.865	0.843	111.83	43.86	77.30	83.26
CMA	0.823	0.919	0.870	0.865	108.63	44.84	77.19	80.61
MedCPT	0.825	0.932	0.872	0.879	106.65	42.05	75.34	77.62
GraphCare	0.825	0.923	0.870	0.872	105.00	42.35	74.26	77.30
GDT	0.878	0.957	0.912	0.919	96.22	38.09	68.79	69.16
SDT (Ours)	0.913	0.968	0.935	0.944	87.12	35.62	63.23	62.63

Table 4: NASA-TLX multi-dimensional workload subscale means (0–100 scale) across all 9 conditions. Lower indicates less workload on all scales except Performance (where higher indicates superior perceived success).

Condition	Mental ↓	Physical ↓	Temporal ↓	Perform. ↑	Effort ↓	Frustr. ↓
Control	66.69	18.10	63.70	56.42	64.81	39.89
BM25	67.61	18.53	64.59	55.57	65.71	40.65
LLM-RAG	52.55	15.44	47.82	63.70	50.53	28.22
ColBERT	51.06	14.60	46.64	65.15	49.14	27.15
CMA	52.76	15.13	48.65	64.62	51.02	28.44
MedCPT	51.02	14.48	46.54	65.53	49.18	27.41
GraphCare	49.35	14.25	45.23	66.46	47.36	25.98
GDT	46.17	13.07	41.49	68.45	44.34	23.28
SDT (Ours)	38.34	11.59	33.12	72.76	36.56	17.47

Table 5: Emergency Department operational impact projected via Erlang C queueing model ($c = 8$ physicians, $\lambda = 6.0 \text{ hr}^{-1}$, baseline $\mu = 0.80 \text{ hr}^{-1}$).

Condition	Δt (s)	μ' (hr^{-1})	ρ'	P_W	W_q (min)
Control	0.0	0.8000	0.9375	0.8073	121.10
BM25	0.0	0.8000	0.9375	0.8073	121.10
LLM-RAG	0.0	0.8000	0.9375	0.8073	121.10
ColBERT	8.6	0.8015	0.9357	0.8019	116.74
CMA	9.8	0.8017	0.9354	0.8011	116.14
MedCPT	12.2	0.8022	0.9349	0.7997	114.97
GraphCare	12.9	0.8023	0.9348	0.7993	114.64
GDT	19.6	0.8035	0.9334	0.7952	111.53
SDT	25.6	0.8046	0.9322	0.7915	108.79

Figure Legends

Figure 1: Mean Time-to-Correct-Information (TTCI) across 9 experimental conditions. Grouped bar chart showing mean TTCI in seconds $\pm$ standard error across $N = 1,000$ vignettes. SDT achieves 62.97 s, reducing search time by 28.9% relative to Control (88.52 s) and 8.7% relative to GDT (68.96 s, paired t -test $p < 10^{-12}$).

Figure 2: Diagnostic retrieval accuracy across conditions. Proportion of diagnostic vignettes successfully resolved within query budget ($Q_{\max} = 5$). SDT resolves 93.9% of cases, outperforming GDT (91.5%), MedCPT (87.5%), GraphCare (87.1%), and Control (86.8%).

Figure 3: Modeled cognitive load across conditions. Cumulative Sweller cognitive load scores. SDT incurs the lowest cognitive burden (34.98), reflecting the efficacy of Sweller-constrained prefetching and Fiedler context attenuation.

Figure 4: Diagnostic accuracy stratified by case complexity. Accuracy comparison between Low Complexity ($n = 469$) and High Complexity ($n = 531$) vignettes. In complex cases with diagnostic pivots, SDT’s advantage expands to +9.0 pp over Control (91.3% vs. 82.3%).

Figure 5: Multi-dimensional NASA-TLX subjective workload profile. Radar chart displaying mean ratings across all six NASA-TLX dimensions (0–100 scale) for representative conditions. SDT (crimson polygon) achieves the lowest workload footprint across Mental Demand, Temporal Demand, Effort, and Frustration, while maximizing perceived Performance.

Figure 6: Emergency Department operational queue waiting time (W_q). Erlang C queueing projections ($c = 8$ physicians, $\lambda = 6.0 \text{ hr}^{-1}$, baseline $\mu = 0.80 \text{ hr}^{-1}$). Saving 25.6 seconds per encounter yields a 12.31-minute reduction in average patient queue wait time, demonstrating super-linear operational leverage near queue saturation.

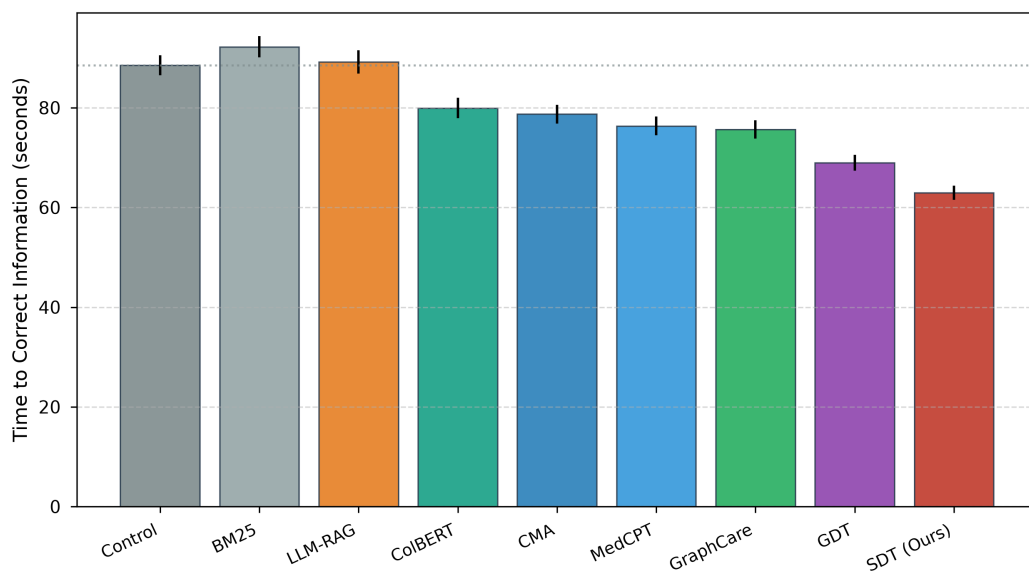


Figure 1: Mean Time-to-Correct-Information (TTCI) across 9 experimental conditions.

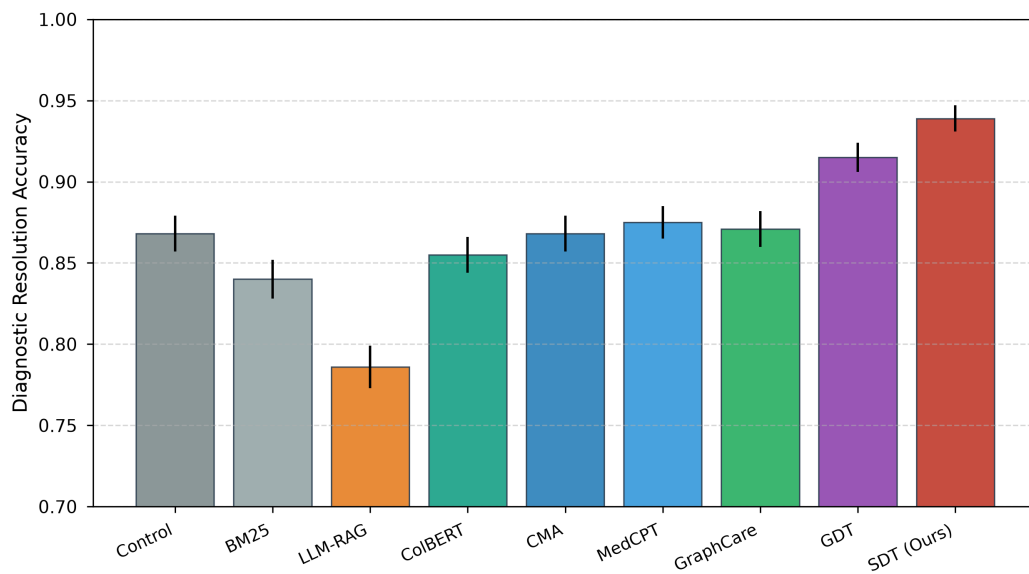


Figure 2: Diagnostic retrieval accuracy across conditions.

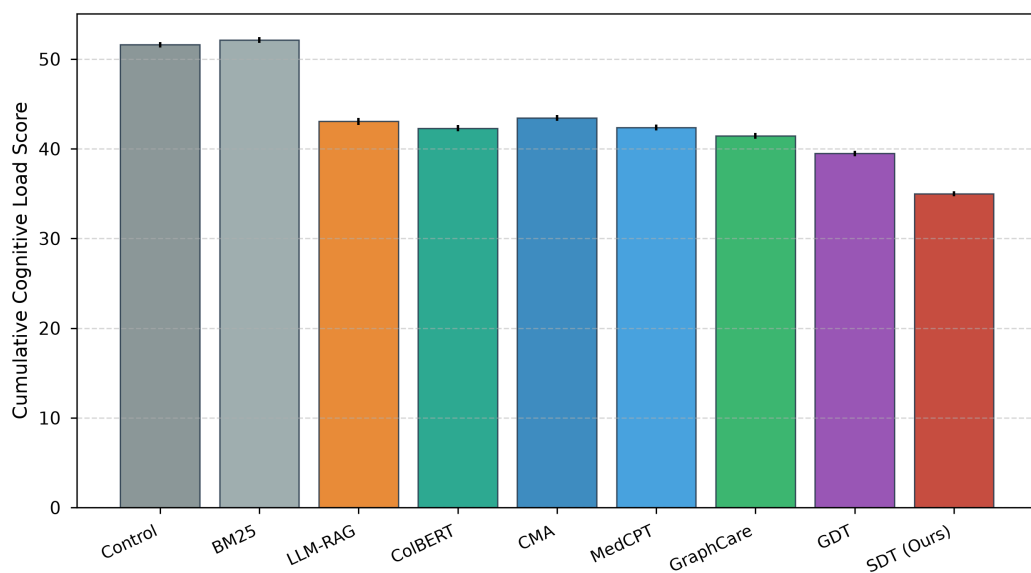


Figure 3: Modeled cognitive load across conditions.

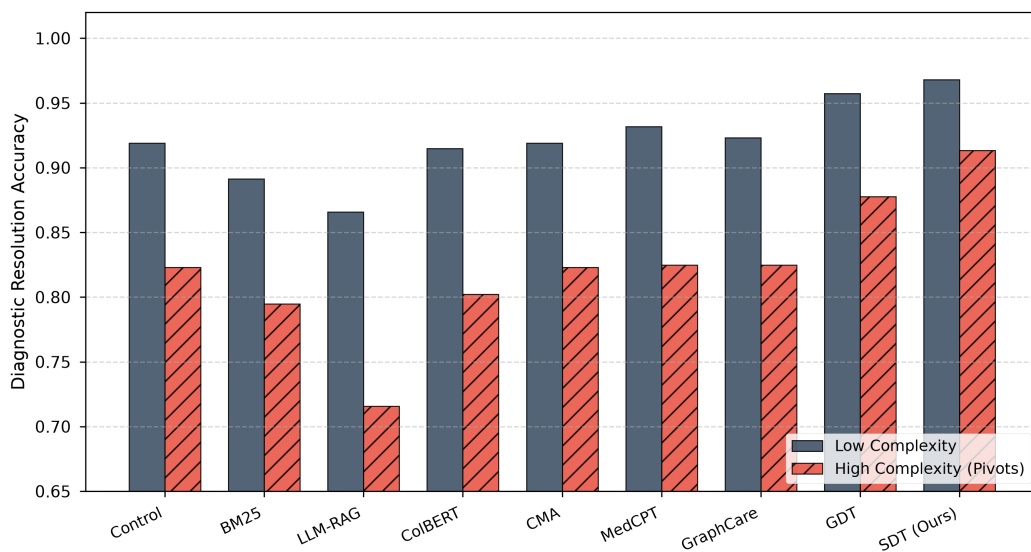


Figure 4: Diagnostic accuracy stratified by case complexity.

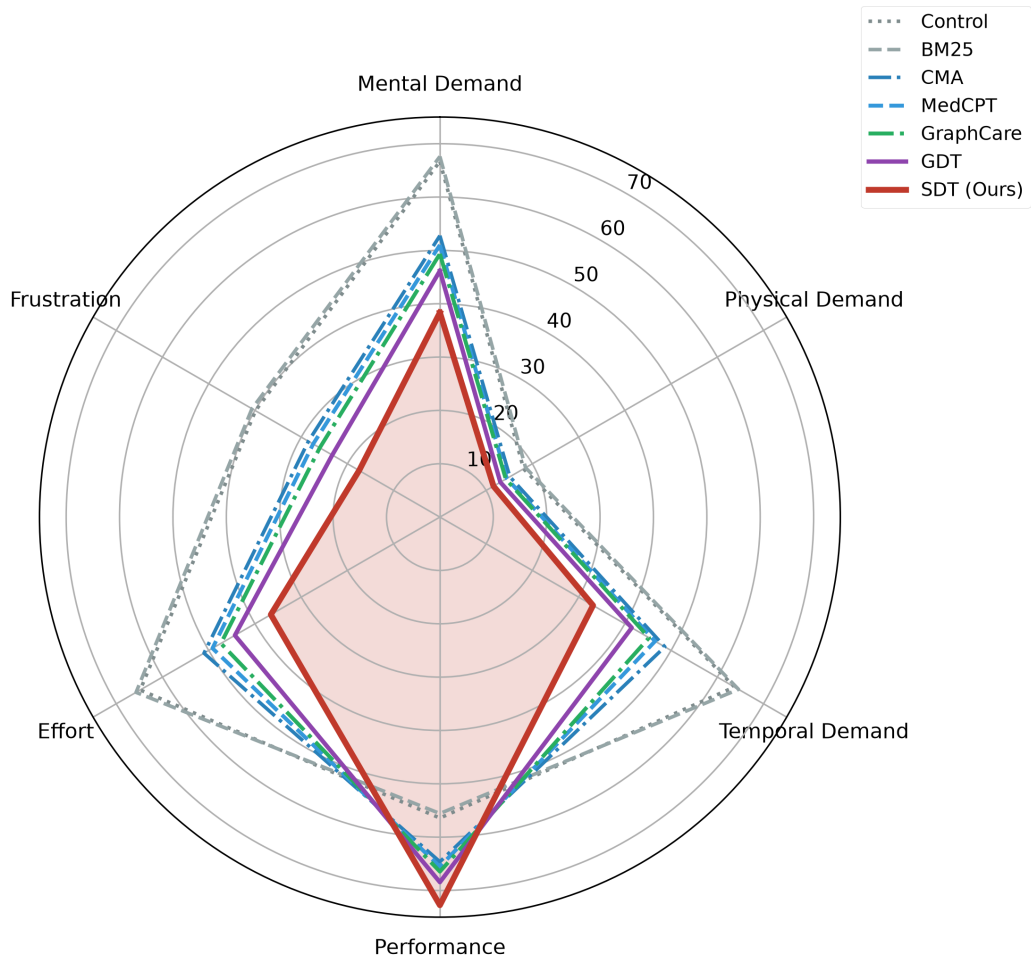


Figure 5: Multi-dimensional NASA-TLX subjective workload profile.

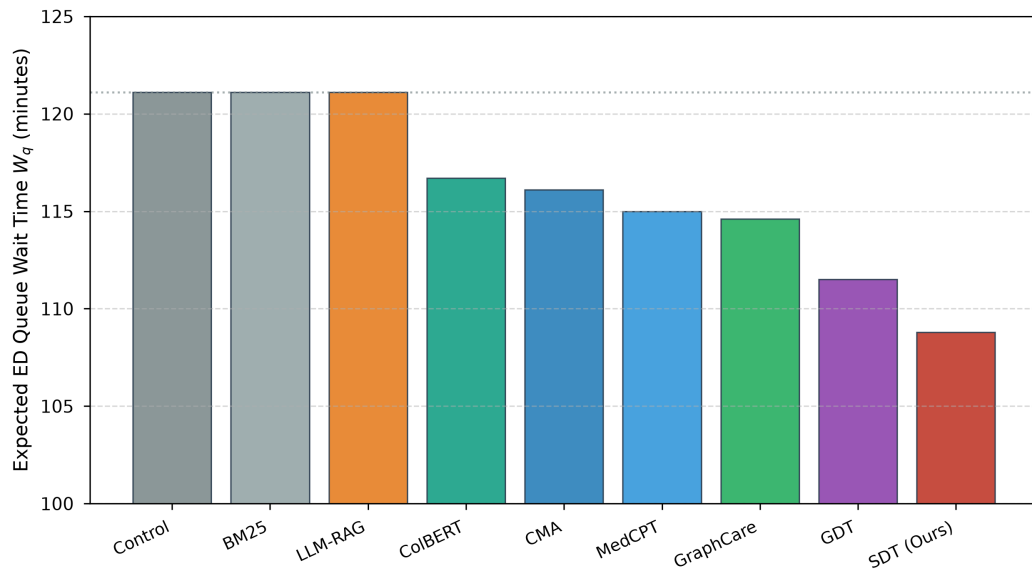


Figure 6: Emergency Department operational queue waiting time (W_q).